The Economic Consequence of Democratic Resilience

The aftermath of the insurrection attempt in South Korea

Byunghwan Son (bson3@gmu.edu)

George Mason University

2026-03-22

The Standard Story — and What It Misses

- Political instability is costly: coups, electoral surprises, regime change ⇒ lower investment, higher volatility, depressed asset prices (Grier, Grier, & Moncrieff 2025; Bernhard & Leblang 2006)
- Democratic backsliding in particular weakens institutional constraints ⇒ reduced policy predictability, higher political risk premium (Haggard & Kaufman 2021; Yoo 2025)
- **But**: what happens when an authoritarian challenge is swiftly and decisively *reversed*?
- If markets price *uncertainty* rather than instability per se, then credible resolution should generate *positive* market responses
- This scenario is underexplored — both theoretically and empirically

The Standard Story — and What It Misses

- Political instability is costly: coups, electoral surprises, regime change ⇒ lower investment, higher volatility, depressed asset prices (Grier, Grier, & Moncrieff 2025; Bernhard & Leblang 2006)
- Democratic backsliding in particular weakens institutional constraints ⇒ reduced policy predictability, higher political risk premium (Haggard & Kaufman 2021; Yoo 2025)
- **But**: what happens when an authoritarian challenge is swiftly and decisively *reversed*?
- If markets price *uncertainty* rather than instability per se, then credible resolution should generate *positive* market responses
- This scenario is underexplored — both theoretically and empirically

The Standard Story — and What It Misses

- Political instability is costly: coups, electoral surprises, regime change ⇒ lower investment, higher volatility, depressed asset prices (Grier, Grier, & Moncrieff 2025; Bernhard & Leblang 2006)
- Democratic backsliding in particular weakens institutional constraints ⇒ reduced policy predictability, higher political risk premium (Haggard & Kaufman 2021; Yoo 2025)
- **But**: what happens when an authoritarian challenge is swiftly and decisively *reversed*?
- If markets price *uncertainty* rather than instability per se, then credible resolution should generate *positive* market responses
- This scenario is underexplored — both theoretically and empirically

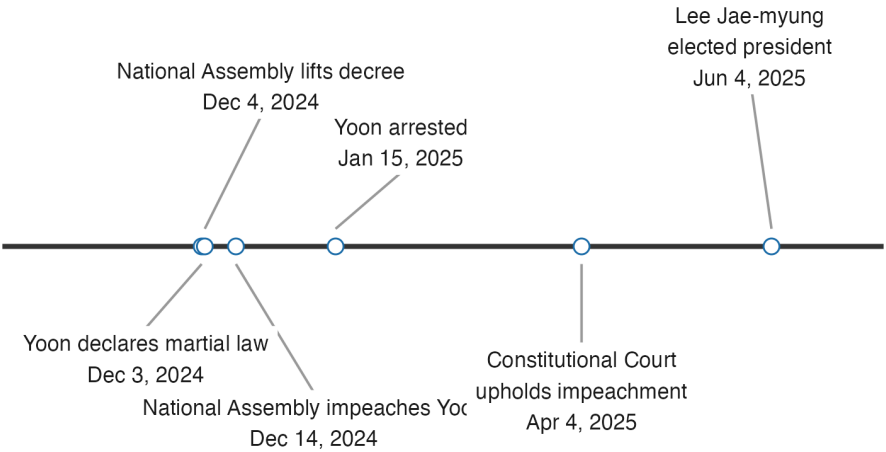
The Standard Story — and What It Misses

- Political instability is costly: coups, electoral surprises, regime change ⇒ lower investment, higher volatility, depressed asset prices (Grier, Grier, & Moncrieff 2025; Bernhard & Leblang 2006)
- Democratic backsliding in particular weakens institutional constraints ⇒ reduced policy predictability, higher political risk premium (Haggard & Kaufman 2021; Yoo 2025)
- **But**: what happens when an authoritarian challenge is swiftly and decisively *reversed*?
- If markets price *uncertainty* rather than instability per se, then credible resolution should generate *positive* market responses
- This scenario is underexplored — both theoretically and empirically

The Standard Story — and What It Misses

- Political instability is costly: coups, electoral surprises, regime change ⇒ lower investment, higher volatility, depressed asset prices (Grier, Grier, & Moncrieff 2025; Bernhard & Leblang 2006)
- Democratic backsliding in particular weakens institutional constraints ⇒ reduced policy predictability, higher political risk premium (Haggard & Kaufman 2021; Yoo 2025)
- **But**: what happens when an authoritarian challenge is swiftly and decisively *reversed*?
- If markets price *uncertainty* rather than instability per se, then credible resolution should generate *positive* market responses
- This scenario is underexplored — both theoretically and empirically

The South Korean Case



Argument: The Democratic Resilience Premium

- Markets incorporate political information because political actors shape the institutional environment for economic activity (North 1990)
- The relevant variable is not instability per se but **uncertainty** — and uncertainty has a resolution as well as an onset
- If a crisis is resolved through credible institutional mechanisms, investors update expectations about long-run governance quality
- Yoon's presidency generated a persistent political risk premium: regulatory confrontation, governance risk, and the "Korea Discount"
- December 3 set in motion a **political process** — not just a shock — that permanently eliminated that premium
- Prediction: initially negative (uncertainty spike), then persistently positive (revaluation) as resolution became certain

Argument: The Democratic Resilience Premium

- Markets incorporate political information because political actors shape the institutional environment for economic activity (North 1990)
- The relevant variable is not instability per se but **uncertainty** — and uncertainty has a resolution as well as an onset
- If a crisis is resolved through credible institutional mechanisms, investors update expectations about long-run governance quality
- Yoon's presidency generated a persistent political risk premium: regulatory confrontation, governance risk, and the "Korea Discount"
- December 3 set in motion a **political process** — not just a shock — that permanently eliminated that premium
- Prediction: initially negative (uncertainty spike), then persistently positive (revaluation) as resolution became certain

Argument: The Democratic Resilience Premium

- Markets incorporate political information because political actors shape the institutional environment for economic activity (North 1990)
- The relevant variable is not instability per se but **uncertainty** — and uncertainty has a resolution as well as an onset
- If a crisis is resolved through credible institutional mechanisms, investors update expectations about long-run governance quality
- Yoon's presidency generated a persistent political risk premium: regulatory confrontation, governance risk, and the "Korea Discount"
- December 3 set in motion a **political process** — not just a shock — that permanently eliminated that premium
- Prediction: initially negative (uncertainty spike), then persistently positive (revaluation) as resolution became certain

Argument: The Democratic Resilience Premium

- Markets incorporate political information because political actors shape the institutional environment for economic activity (North 1990)
- The relevant variable is not instability per se but **uncertainty** — and uncertainty has a resolution as well as an onset
- If a crisis is resolved through credible institutional mechanisms, investors update expectations about long-run governance quality
- Yoon's presidency generated a persistent political risk premium: regulatory confrontation, governance risk, and the "Korea Discount"
- December 3 set in motion a **political process** — not just a shock — that permanently eliminated that premium
- Prediction: initially negative (uncertainty spike), then persistently positive (revaluation) as resolution became certain

Argument: The Democratic Resilience Premium

- Markets incorporate political information because political actors shape the institutional environment for economic activity (North 1990)
- The relevant variable is not instability per se but **uncertainty** — and uncertainty has a resolution as well as an onset
- If a crisis is resolved through credible institutional mechanisms, investors update expectations about long-run governance quality
- Yoon's presidency generated a persistent political risk premium: regulatory confrontation, governance risk, and the "Korea Discount"
- December 3 set in motion a **political process** — not just a shock — that permanently eliminated that premium
- Prediction: initially negative (uncertainty spike), then persistently positive (revaluation) as resolution became certain

Argument: The Democratic Resilience Premium

- Markets incorporate political information because political actors shape the institutional environment for economic activity (North 1990)
- The relevant variable is not instability per se but **uncertainty** — and uncertainty has a resolution as well as an onset
- If a crisis is resolved through credible institutional mechanisms, investors update expectations about long-run governance quality
- Yoon's presidency generated a persistent political risk premium: regulatory confrontation, governance risk, and the "Korea Discount"
- December 3 set in motion a **political process** — not just a shock — that permanently eliminated that premium
- Prediction: initially negative (uncertainty spike), then persistently positive (revaluation) as resolution became certain

Hypotheses

H1: Market Revaluation (ASCM)

The political process initiated by the December 3 martial law declaration leads to a positive revaluation of Korean equities.

H1a: Daily News Pricing (EGARCH)

News favorable to Yoon's political survival is associated with **lower subsequent returns** and **higher volatility**.

Why Augmented Synthetic Control (ASCM)?

- Classical SCM (Abadie, Diamond, & Hainmueller 2010) requires:

$$\hat{y}_{0t} = \sum_j w_j y_{jt}, \quad w_j \geq 0, \quad \sum_j w_j = 1$$

- This requires Korea's log price to lie within the **convex hull** of donors — violated by construction (level problem)
- Result: degenerate corner solutions (100% to China, MSCI, or Taiwan) across all classical SCM specifications
- **ASCM** (Ben-Michael, Feller & Rothstein 2021) resolves this via:
  - Ridge penalty on weights (shrinks toward equal weights)
  - Intercept μ_0 (allows weights to sum to 0 and absorb level differences, rather than forcing the synthetic unit to match levels)
  - Ridge penalty on the outcome model (prevents overfitting when the number of donors approaches the number of pre-treatment periods)

Why Augmented Synthetic Control (ASCM)?

- Classical SCM (Abadie, Diamond, & Hainmueller 2010) requires:

$$\hat{y}_{0t} = \sum_j w_j y_{jt}, \quad w_j \geq 0, \quad \sum_j w_j = 1$$

- This requires Korea's log price to lie within the **convex hull** of donors — violated by construction (level problem)
- Result: degenerate corner solutions (100% to China, MSCI, or Taiwan) across all classical SCM specifications
- **ASCM** (Ben-Michael, Feller & Rothstein 2021) resolves this via:
  - Ridge bias correction with data-driven penalty
  - Allowing negative weights (extrapolation) when pre-period fit requires it, with penalty shrinking toward zero
  - Jackknife inference

Why Augmented Synthetic Control (ASCM)?

- Classical SCM (Abadie, Diamond, & Hainmueller 2010) requires:

$$\hat{y}_{0t} = \sum_j w_j y_{jt}, \quad w_j \geq 0, \quad \sum_j w_j = 1$$

- This requires Korea's log price to lie within the **convex hull** of donors — violated by construction (level problem)
- Result: degenerate corner solutions (100% to China, MSCI, or Taiwan) across all classical SCM specifications
- **ASCM** (Ben-Michael, Feller & Rothstein 2021) resolves this via:
  1. Unit fixed effects absorb arbitrary level differences
  2. Ridge penalty shrinks weights toward uniform when donor fit is poor, preventing corner solutions
  3. Jackknife bias correction adjusts for residual imbalance

Why Augmented Synthetic Control (ASCM)?

- Classical SCM (Abadie, Diamond, & Hainmueller 2010) requires:

$$\hat{y}_{0t} = \sum_j w_j y_{jt}, \quad w_j \geq 0, \quad \sum_j w_j = 1$$

- This requires Korea's log price to lie within the **convex hull** of donors — violated by construction (level problem)
- Result: degenerate corner solutions (100% to China, MSCI, or Taiwan) across all classical SCM specifications
- **ASCM** (Ben-Michael, Feller & Rothstein 2021) resolves this via:
  1. Unit fixed effects absorb arbitrary level differences
  2. Ridge regression bias correction adjusts for residual imbalance *without* requiring the convex hull condition
  3. Permits **negative donor weights** — necessary when counterfactual lies outside the convex hull

Why Augmented Synthetic Control (ASCM)?

- Classical SCM (Abadie, Diamond, & Hainmueller 2010) requires:

$$\hat{y}_{0t} = \sum_j w_j y_{jt}, \quad w_j \geq 0, \quad \sum_j w_j = 1$$

- This requires Korea's log price to lie within the **convex hull** of donors — violated by construction (level problem)
- Result: degenerate corner solutions (100% to China, MSCI, or Taiwan) across all classical SCM specifications
- **ASCM** (Ben-Michael, Feller & Rothstein 2021) resolves this via:
  1. Unit fixed effects absorb arbitrary level differences
  2. Ridge regression bias correction adjusts for residual imbalance *without* requiring the convex hull condition
  3. Permits **negative donor weights** — necessary when counterfactual lies outside the convex hull

Why Augmented Synthetic Control (ASCM)?

- Classical SCM (Abadie, Diamond, & Hainmueller 2010) requires:

$$\hat{y}_{0t} = \sum_j w_j y_{jt}, \quad w_j \geq 0, \quad \sum_j w_j = 1$$

- This requires Korea's log price to lie within the **convex hull** of donors — violated by construction (level problem)
- Result: degenerate corner solutions (100% to China, MSCI, or Taiwan) across all classical SCM specifications
- **ASCM** (Ben-Michael, Feller & Rothstein 2021) resolves this via:
  1. Unit fixed effects absorb arbitrary level differences
  2. Ridge regression bias correction adjusts for residual imbalance *without* requiring the convex hull condition
  3. Permits **negative donor weights** — necessary when counterfactual lies outside the convex hull

Why Augmented Synthetic Control (ASCM)?

- Classical SCM (Abadie, Diamond, & Hainmueller 2010) requires:

$$\hat{y}_{0t} = \sum_j w_j y_{jt}, \quad w_j \geq 0, \quad \sum_j w_j = 1$$

- This requires Korea's log price to lie within the **convex hull** of donors — violated by construction (level problem)
- Result: degenerate corner solutions (100% to China, MSCI, or Taiwan) across all classical SCM specifications
- **ASCM** (Ben-Michael, Feller & Rothstein 2021) resolves this via:
  1. Unit fixed effects absorb arbitrary level differences
  2. Ridge regression bias correction adjusts for residual imbalance *without* requiring the convex hull condition
  3. Permits **negative donor weights** — necessary when counterfactual lies outside the convex hull

Stock Market Data

- Outcome$_1$ (ASCM): daily **log closing price**

$$y_{it} = \ln(P_{it})$$

  where P_{it} is the Yahoo Finance `Close` price for country i on day t
- Treated unit: South Korea (KOSPI Composite Index)
- Pre-treatment: March 4 – December 3, 2024 (113 trading days)
- Post-treatment: December 4, 2024 – end of sample (207 trading days)
- Donor pool: Japan, Taiwan, Hong Kong, Singapore, China, Australia, MSCI Emerging Markets ETF
- Outcome$_2$ (EGARCH): daily **log return**

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Stock Market Data

- Outcome$_1$ (ASCM): daily **log closing price**

$$y_{it} = \ln(P_{it})$$

  where P_{it} is the Yahoo Finance `Close` price for country i on day t
- Treated unit: South Korea (KOSPI Composite Index)
- Pre-treatment: March 4 – December 3, 2024 (113 trading days)
- Post-treatment: December 4, 2024 – end of sample (207 trading days)
- Donor pool: Japan, Taiwan, Hong Kong, Singapore, China, Australia, MSCI Emerging Markets ETF
- Outcome$_2$ (EGARCH): daily **log return**

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Stock Market Data

- Outcome$_1$ (ASCM): daily **log closing price**

$$y_{it} = \ln(P_{it})$$

  where P_{it} is the Yahoo Finance `Close` price for country i on day t
- Treated unit: South Korea (KOSPI Composite Index)
- Pre-treatment: March 4 – December 3, 2024 (113 trading days)
- Post-treatment: December 4, 2024 – end of sample (207 trading days)
- Donor pool: Japan, Taiwan, Hong Kong, Singapore, China, Australia, MSCI Emerging Markets ETF
- Outcome$_2$ (EGARCH): daily **log return**

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Stock Market Data

- Outcome$_1$ (ASCM): daily **log closing price**

$$y_{it} = \ln(P_{it})$$

  where P_{it} is the Yahoo Finance `Close` price for country i on day t
- Treated unit: South Korea (KOSPI Composite Index)
- Pre-treatment: March 4 – December 3, 2024 (113 trading days)
- Post-treatment: December 4, 2024 – end of sample (207 trading days)
- Donor pool: Japan, Taiwan, Hong Kong, Singapore, China, Australia, MSCI Emerging Markets ETF
- Outcome$_2$ (EGARCH): daily **log return**

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Stock Market Data

- Outcome$_1$ (ASCM): daily **log closing price**

$$y_{it} = \ln(P_{it})$$

  where P_{it} is the Yahoo Finance `Close` price for country i on day t
- Treated unit: South Korea (KOSPI Composite Index)
- Pre-treatment: March 4 – December 3, 2024 (113 trading days)
- Post-treatment: December 4, 2024 – end of sample (207 trading days)
- Donor pool: Japan, Taiwan, Hong Kong, Singapore, China, Australia, MSCI Emerging Markets ETF
- Outcome$_2$ (EGARCH): daily **log return**

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Stock Market Data

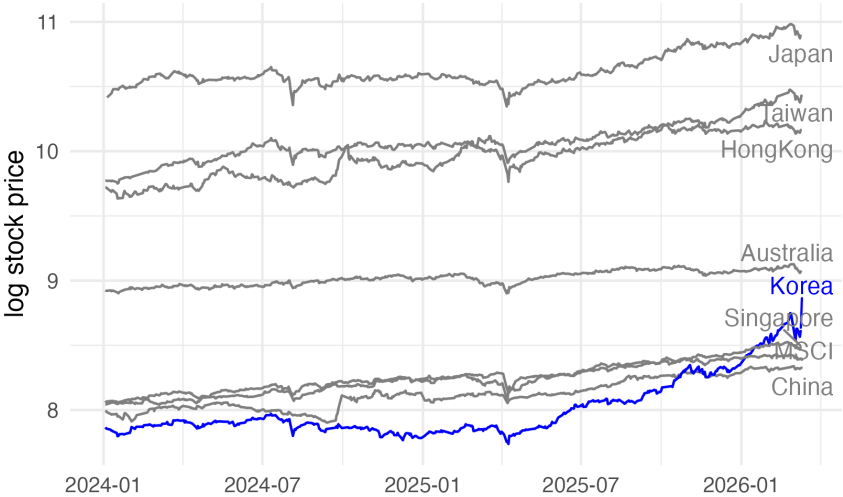
- Outcome$_1$ (ASCM): daily **log closing price**

$$y_{it} = \ln(P_{it})$$

  where P_{it} is the Yahoo Finance `Close` price for country i on day t
- Treated unit: South Korea (KOSPI Composite Index)
- Pre-treatment: March 4 – December 3, 2024 (113 trading days)
- Post-treatment: December 4, 2024 – end of sample (207 trading days)
- Donor pool: Japan, Taiwan, Hong Kong, Singapore, China, Australia, MSCI Emerging Markets ETF
- Outcome$_2$ (EGARCH): daily **log return**

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Why These Donors? The Level Problem (ASCM)



ASCM Estimator and Inference

Estimator

$$\hat{\tau}_t = y_{1t} - \left(\sum_j \hat{w}_j \, y_{jt} + \hat{m}(x_t) \right)$$

where $\hat{w}_j$ are synthetic control weights and $\hat{m}(\cdot)$ is the ridge bias correction term

- Ridge penalty λ selected by cross-validation; optimal $\lambda = 0.00395$ (very small — weights do most of the work)
- **Inference**: jackknife+ (Barber et al., 2021) — valid with $N_1 = 1$ by leaving out one donor at a time
- Avoids cross-unit permutation requirement (undefined when $N_1 = 1$)

ASCM Estimator and Inference

Estimator

$$\hat{\tau}_t = y_{1t} - \left(\sum_j \hat{w}_j \, y_{jt} + \hat{m}(x_t) \right)$$

where $\hat{w}_j$ are synthetic control weights and $\hat{m}(\cdot)$ is the ridge bias correction term

- Ridge penalty λ selected by cross-validation; optimal $\lambda = 0.00395$ (very small — weights do most of the work)
- **Inference**: jackknife+ (Barber et al., 2021) — valid with $N_1 = 1$ by leaving out one donor at a time
- Avoids cross-unit permutation requirement (undefined when $N_1 = 1$)

ASCM Estimator and Inference

Estimator

$$\hat{\tau}_t = y_{1t} - \left(\sum_j \hat{w}_j \, y_{jt} + \hat{m}(x_t) \right)$$

where $\hat{w}_j$ are synthetic control weights and $\hat{m}(\cdot)$ is the ridge bias correction term

- Ridge penalty λ selected by cross-validation; optimal $\lambda = 0.00395$ (very small — weights do most of the work)
- **Inference**: jackknife+ (Barber et al., 2021) — valid with $N_1 = 1$ by leaving out one donor at a time
- Avoids cross-unit permutation requirement (undefined when $N_1 = 1$)

Donor Weights

Country	Weight
Australia	$+1.356$
Japan	$+0.823$
Taiwan	$+0.481$
HongKong	$+0.078$
China	-0.292
MSCI	-0.646
Singapore	-0.800

Weights sum to 1.0

- Australia $> 100\%$ — extrapolation outside convex hull
- Negative weights: Singapore, MSCI, China trending upward in ways Korea was not; ridge correction subtracts them
- **Not a sign of model failure** — the mechanism by which ridge achieves pre-treatment balance
- LOO robustness: full-period ATT positive across all 7 LOO specifications

Donor Weights

Country	Weight
Australia	$+1.356$
Japan	$+0.823$
Taiwan	$+0.481$
HongKong	$+0.078$
China	-0.292
MSCI	-0.646
Singapore	-0.800

Weights sum to 1.0

- Australia $> 100\%$ — extrapolation outside convex hull
- Negative weights: Singapore, MSCI, China trending upward in ways Korea was not; ridge correction subtracts them
- **Not a sign of model failure** — the mechanism by which ridge achieves pre-treatment balance
- LOO robustness: full-period ATT positive across all 7 LOO specifications

Donor Weights

Country	Weight
Australia	$+1.356$
Japan	$+0.823$
Taiwan	$+0.481$
HongKong	$+0.078$
China	-0.292
MSCI	-0.646
Singapore	-0.800

Weights sum to 1.0

- Australia $> 100\%$ — extrapolation outside convex hull
- Negative weights: Singapore, MSCI, China trending upward in ways Korea was not; ridge correction subtracts them
- **Not a sign of model failure** — the mechanism by which ridge achieves pre-treatment balance
- LOO robustness: full-period ATT positive across all 7 LOO specifications

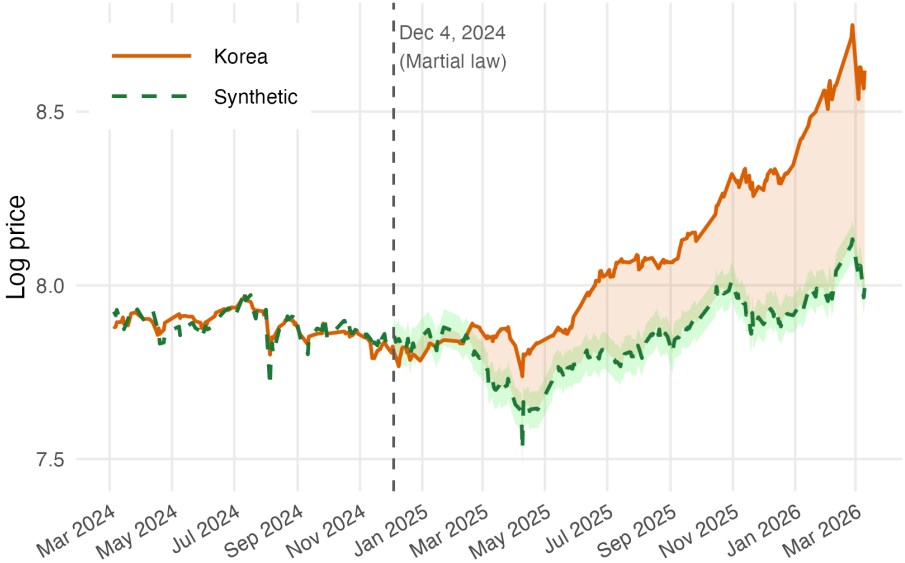
Donor Weights

Country	Weight
Australia	$+1.356$
Japan	$+0.823$
Taiwan	$+0.481$
HongKong	$+0.078$
China	-0.292
MSCI	-0.646
Singapore	-0.800

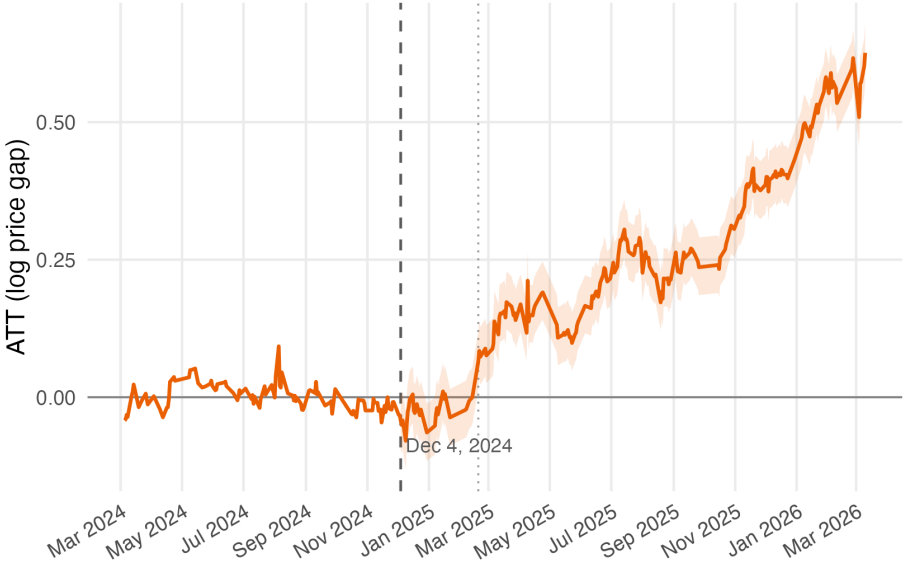
Weights sum to 1.0

- Australia $> 100\%$ — extrapolation outside convex hull
- Negative weights: Singapore, MSCI, China trending upward in ways Korea was not; ridge correction subtracts them
- **Not a sign of model failure** — the mechanism by which ridge achieves pre-treatment balance
- LOO robustness: full-period ATT positive across all 7 LOO specifications

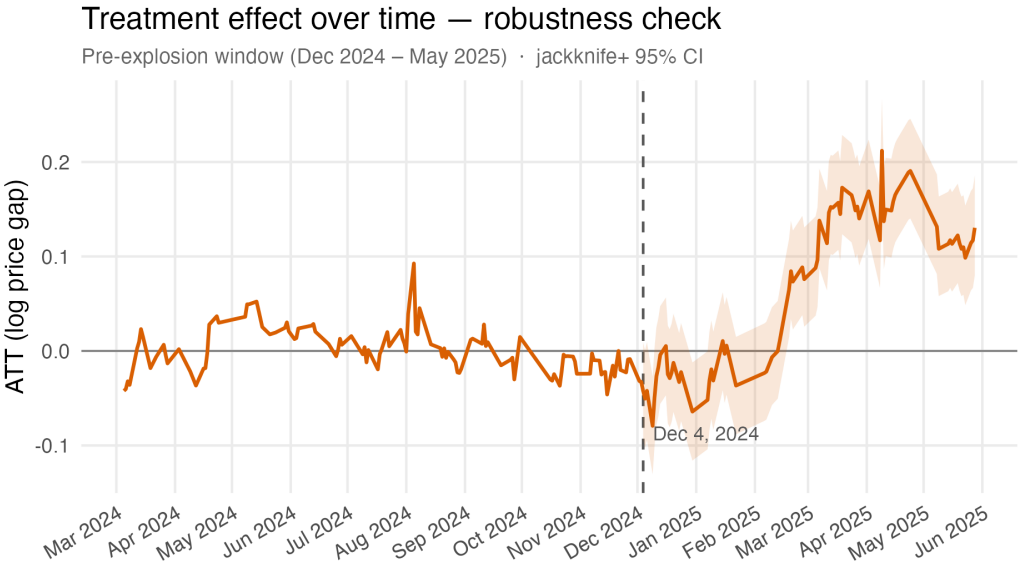
Main Result: Path Plot (ATT $\approx$ 24%)



Main Result: Gap Plot (ATT $\approx$ 24%)



Robustness: Pre-Explosion Window



Political News Data (EGARCH)

- 9,427 Korean- and English-language news headlines, August 2024 – March 2026
- Collected via Google News RSS API; search terms: "Yoon Suk-yeol" and "Lee Jae-myung" (and Korean equivalents)
- Sources: Hankyoreh, Joongang Ilbo, Kyunghyang Sinmun, Chosun Ilbo, Donga Ilbo, NYT, WaPo, FT
- **Classification**: OpenAI ChatGPT API prompted to assign each headline to {very bad, bad, neutral, good, very good} with respect to Yoon's *political survival*
- Daily measure: sum of scores (-2 to $+2$) over all headlines on that day
- **Validation**: 400-headline hand-coded sample; agreement rate $\approx 89\%$ vs. $\approx 72\%$ for conventional ML classifier

Political News Data (EGARCH)

- 9,427 Korean- and English-language news headlines, August 2024 – March 2026
- Collected via Google News RSS API; search terms: "Yoon Suk-yeol" and "Lee Jae-myung" (and Korean equivalents)
- Sources: Hankyoreh, Joongang Ilbo, Kyunghyang Sinmun, Chosun Ilbo, Donga Ilbo, NYT, WaPo, FT
- **Classification**: OpenAI ChatGPT API prompted to assign each headline to {very bad, bad, neutral, good, very good} with respect to Yoon's *political survival*
- Daily measure: sum of scores (-2 to $+2$) over all headlines on that day
- **Validation**: 400-headline hand-coded sample; agreement rate $\approx 89\%$ vs. $\approx 72\%$ for conventional ML classifier

Political News Data (EGARCH)

- 9,427 Korean- and English-language news headlines, August 2024 – March 2026
- Collected via Google News RSS API; search terms: "Yoon Suk-yeol" and "Lee Jae-myung" (and Korean equivalents)
- Sources: Hankyoreh, Joongang Ilbo, Kyunghyang Sinmun, Chosun Ilbo, Donga Ilbo, NYT, WaPo, FT
- **Classification**: OpenAI ChatGPT API prompted to assign each headline to {very bad, bad, neutral, good, very good} with respect to Yoon's *political survival*
- Daily measure: sum of scores (-2 to $+2$) over all headlines on that day
- **Validation**: 400-headline hand-coded sample; agreement rate $\approx 89\%$ vs. $\approx 72\%$ for conventional ML classifier

Political News Data (EGARCH)

- 9,427 Korean- and English-language news headlines, August 2024 – March 2026
- Collected via Google News RSS API; search terms: "Yoon Suk-yeol" and "Lee Jae-myung" (and Korean equivalents)
- Sources: Hankyoreh, Joongang Ilbo, Kyunghyang Sinmun, Chosun Ilbo, Donga Ilbo, NYT, WaPo, FT
- **Classification**: OpenAI ChatGPT API prompted to assign each headline to {very bad, bad, neutral, good, very good} with respect to Yoon's *political survival*
- Daily measure: sum of scores (-2 to $+2$) over all headlines on that day
- **Validation**: 400-headline hand-coded sample; agreement rate $\approx 89\%$ vs. $\approx 72\%$ for conventional ML classifier

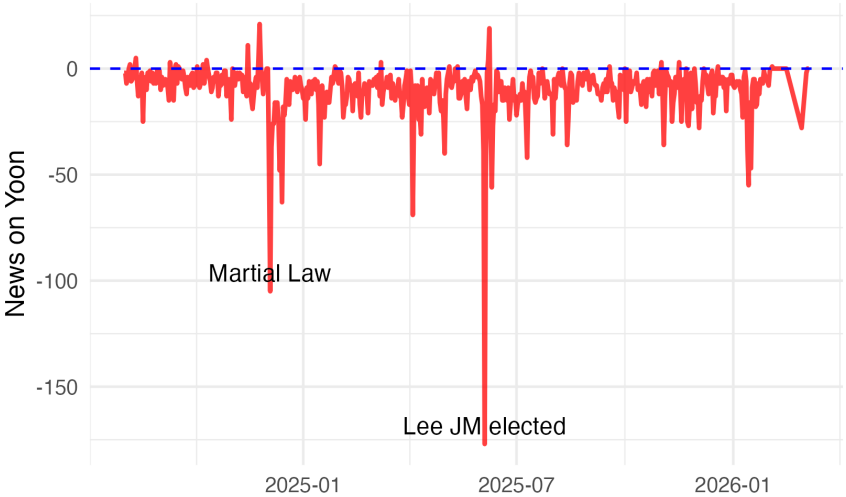
Political News Data (EGARCH)

- 9,427 Korean- and English-language news headlines, August 2024 – March 2026
- Collected via Google News RSS API; search terms: "Yoon Suk-yeol" and "Lee Jae-myung" (and Korean equivalents)
- Sources: Hankyoreh, Joongang Ilbo, Kyunghyang Sinmun, Chosun Ilbo, Donga Ilbo, NYT, WaPo, FT
- **Classification**: OpenAI ChatGPT API prompted to assign each headline to {very bad, bad, neutral, good, very good} with respect to Yoon's *political survival*
- Daily measure: sum of scores (-2 to $+2$) over all headlines on that day
- **Validation**: 400-headline hand-coded sample; agreement rate $\approx 89\%$ vs. $\approx 72\%$ for conventional ML classifier

Political News Data (EGARCH)

- 9,427 Korean- and English-language news headlines, August 2024 – March 2026
- Collected via Google News RSS API; search terms: "Yoon Suk-yeol" and "Lee Jae-myung" (and Korean equivalents)
- Sources: Hankyoreh, Joongang Ilbo, Kyunghyang Sinmun, Chosun Ilbo, Donga Ilbo, NYT, WaPo, FT
- **Classification**: OpenAI ChatGPT API prompted to assign each headline to {very bad, bad, neutral, good, very good} with respect to Yoon's *political survival*
- Daily measure: sum of scores (-2 to $+2$) over all headlines on that day
- **Validation**: 400-headline hand-coded sample; agreement rate $\approx 89\%$ vs. $\approx 72\%$ for conventional ML classifier

Political News Trend



EGARCH Model

Mean equation

$$r_t = \mu + \phi_1 r_{t-1} + \delta\hat{\sigma}_t + \beta_1 \text{YoonNews}_{t-1} + \beta_2 \text{YoonNews}_t + \beta_3 \text{USUncertainty}_{t-1} + \varepsilon_t$$

Variance equation

$$\ln \sigma_t^2 = \omega + \alpha_1(|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma_1 z_{t-1} + \beta_1^v \ln \sigma_{t-1}^2 + [...]$$

- $\varepsilon_t = \sigma_t z_t$, $z_t \sim \text{std}(\nu)$
- $\gamma_1 > 0$: leverage effect (negative shocks $\uparrow$ volatility more)
- YoonNews_{t-1}: primary variable of interest (lagged — market reaction the following day)
- YoonNews_t: same-day market correction

EGARCH Model

Mean equation

$$r_t = \mu + \phi_1 r_{t-1} + \delta\hat{\sigma}_t + \beta_1 \text{YoonNews}_{t-1} + \beta_2 \text{YoonNews}_t + \beta_3 \text{USUncertainty}_{t-1} + \varepsilon_t$$

Variance equation

$$\ln \sigma_t^2 = \omega + \alpha_1(|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma_1 z_{t-1} + \beta_1^v \ln \sigma_{t-1}^2 + [...]$$

- $\varepsilon_t = \sigma_t z_t$, $z_t \sim \text{std}(\nu)$
- $\gamma_1 > 0$: leverage effect (negative shocks $\uparrow$ volatility more)
- YoonNews_{t-1}: primary variable of interest (lagged — market reaction the following day)
- YoonNews_t: same-day market correction

EGARCH Model

Mean equation

$$r_t = \mu + \phi_1 r_{t-1} + \delta\hat{\sigma}_t + \beta_1 \text{YoonNews}_{t-1} + \beta_2 \text{YoonNews}_t + \beta_3 \text{USUncertainty}_{t-1} + \varepsilon_t$$

Variance equation

$$\ln \sigma_t^2 = \omega + \alpha_1 (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma_1 z_{t-1} + \beta_1^v \ln \sigma_{t-1}^2 + [...]$$

- $\varepsilon_t = \sigma_t z_t$, $z_t \sim \text{std}(\nu)$
- $\gamma_1 > 0$: leverage effect (negative shocks $\uparrow$ volatility more)
- YoonNews_{t-1}: primary variable of interest (lagged — market reaction the following day)
- YoonNews_t: same-day market correction

EGARCH Model

Mean equation

$$r_t = \mu + \phi_1 r_{t-1} + \delta\hat{\sigma}_t + \beta_1 \text{YoonNews}_{t-1} + \beta_2 \text{YoonNews}_t + \beta_3 \text{USUncertainty}_{t-1} + \varepsilon_t$$

Variance equation

$$\ln \sigma_t^2 = \omega + \alpha_1 (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma_1 z_{t-1} + \beta_1^v \ln \sigma_{t-1}^2 + [...]$$

- $\varepsilon_t = \sigma_t z_t$, $z_t \sim \text{std}(\nu)$
- $\gamma_1 > 0$: leverage effect (negative shocks $\uparrow$ volatility more)
- YoonNews_{t-1}: primary variable of interest (lagged — market reaction the following day)
- YoonNews_t: same-day market correction

EGARCH Results: Mean Equation

Parameter	Equation	Estimate	(Rob. SE)	
μ	Mean	-0.01235	(0.00129)	$***$
ar1	Mean	-0.13464	(0.03087)	$***$
archm (δ)	Mean	0.74856	(0.08823)	$***$
YoonNews$_t$ (β_1)	Mean	$+0.00194$	(0.00028)	$***$
YoonNews$_{t-1}$ (β_2)	Mean	-0.00036	(0.00016)	$**$
USUncertainty$_{t-1}$	Mean	-0.00065	(0.00028)	$**$

$^{***}p < 0.01$, $^{**}p < 0.05$. Robust SE.

- $\beta_2 < 0$: Yoon-favorable news depresses next-day returns
- $\beta_1 > 0$: same-day partial correction (buy-low behavior)

EGARCH Results: Mean Equation

Parameter	Equation	Estimate	(Rob. SE)	
μ	Mean	-0.01235	(0.00129)	$***$
ar1	Mean	-0.13464	(0.03087)	$***$
archm (δ)	Mean	0.74856	(0.08823)	$***$
YoonNews$_t$ (β_1)	Mean	$+0.00194$	(0.00028)	$***$
YoonNews$_{t-1}$ (β_2)	Mean	-0.00036	(0.00016)	$**$
USUncertainty$_{t-1}$	Mean	-0.00065	(0.00028)	$**$

$^{***}p < 0.01$, $^{**}p < 0.05$. Robust SE.

- $\beta_2 < 0$: Yoon-favorable news depresses next-day returns
- $\beta_1 > 0$: same-day partial correction (buy-low behavior)

EGARCH Results: Variance Equation

Parameter	Equation	Estimate	(Rob. SE)	
ω	Variance	-3.5597	(0.4326)	$***$
α_1	Variance	0.2636	(0.0643)	$***$
β_1^v	Variance	0.5877	(0.0495)	$***$
γ_1 (leverage)	Variance	0.1299	(0.0408)	$***$
YoonNews$_t$ (θ_1)	Variance	-0.1259	(0.0219)	$***$
YoonNews$_{t-1}$ (θ_2)	Variance	$+0.0956$	(0.0124)	$***$
USUncertainty$_{t-1}$	Variance	-0.0304	(0.0168)	$*$
shape (ν)	Dist.	4.3695	(0.9219)	$***$

LogLik $= 1071.2$. AIC $= -5.793$.

- $\theta_2 > 0$: Yoon-favorable news *raises* next-day volatility
- $\theta_1 < 0$: same-day volatility *falls*; $\gamma_1 > 0$: leverage confirmed

EGARCH Results: Variance Equation

Parameter	Equation	Estimate	(Rob. SE)	
ω	Variance	-3.5597	(0.4326)	$***$
α_1	Variance	0.2636	(0.0643)	$***$
β_1^v	Variance	0.5877	(0.0495)	$***$
γ_1 (leverage)	Variance	0.1299	(0.0408)	$***$
YoonNews$_t$ (θ_1)	Variance	-0.1259	(0.0219)	$***$
YoonNews$_{t-1}$ (θ_2)	Variance	$+0.0956$	(0.0124)	$***$
USUncertainty$_{t-1}$	Variance	-0.0304	(0.0168)	$*$
shape (ν)	Dist.	4.3695	(0.9219)	$***$

LogLik $= 1071.2$. AIC $= -5.793$.

- $\theta_2 > 0$: Yoon-favorable news *raises* next-day volatility
- $\theta_1 < 0$: same-day volatility *falls*; $\gamma_1 > 0$: leverage confirmed

What the South Korean Case Shows

- Markets do not simply respond to political *instability* — they respond to political *uncertainty* and its resolution
- When democratic institutions contain a crisis transparently and irreversibly, investors update expectations about long-run institutional quality
- The four-phase dynamic — shock, uncertainty, turning point, revaluation — emerged from the data and maps precisely onto the political timeline
- **First statistically significant day: February 19, 2025** — when constitutional certainty was established, not at the shock
- **Full-period ATT** $\approx 24\%$ — the capitalized value of removing the Yoon political risk premium
- "Markets noticed, and they rewarded it"

What the South Korean Case Shows

- Markets do not simply respond to political *instability* — they respond to political *uncertainty* and its resolution
- When democratic institutions contain a crisis transparently and irreversibly, investors update expectations about long-run institutional quality
- The four-phase dynamic — shock, uncertainty, turning point, revaluation — emerged from the data and maps precisely onto the political timeline
- **First statistically significant day: February 19, 2025** — when constitutional certainty was established, not at the shock
- **Full-period ATT** $\approx 24\%$ — the capitalized value of removing the Yoon political risk premium
- "Markets noticed, and they rewarded it"

What the South Korean Case Shows

- Markets do not simply respond to political *instability* — they respond to political *uncertainty* and its resolution
- When democratic institutions contain a crisis transparently and irreversibly, investors update expectations about long-run institutional quality
- The four-phase dynamic — shock, uncertainty, turning point, revaluation — emerged from the data and maps precisely onto the political timeline
- **First statistically significant day: February 19, 2025** — when constitutional certainty was established, not at the shock
- **Full-period ATT** $\approx 24\%$ — the capitalized value of removing the Yoon political risk premium
- "Markets noticed, and they rewarded it"

What the South Korean Case Shows

- Markets do not simply respond to political *instability* — they respond to political *uncertainty* and its resolution
- When democratic institutions contain a crisis transparently and irreversibly, investors update expectations about long-run institutional quality
- The four-phase dynamic — shock, uncertainty, turning point, revaluation — emerged from the data and maps precisely onto the political timeline
- **First statistically significant day: February 19, 2025** — when constitutional certainty was established, not at the shock
- **Full-period ATT** $\approx 24\%$ — the capitalized value of removing the Yoon political risk premium
- "Markets noticed, and they rewarded it"

What the South Korean Case Shows

- Markets do not simply respond to political *instability* — they respond to political *uncertainty* and its resolution
- When democratic institutions contain a crisis transparently and irreversibly, investors update expectations about long-run institutional quality
- The four-phase dynamic — shock, uncertainty, turning point, revaluation — emerged from the data and maps precisely onto the political timeline
- **First statistically significant day: February 19, 2025** — when constitutional certainty was established, not at the shock
- **Full-period ATT** $\approx 24\%$ — the capitalized value of removing the Yoon political risk premium
- "Markets noticed, and they rewarded it"

What the South Korean Case Shows

- Markets do not simply respond to political *instability* — they respond to political *uncertainty* and its resolution
- When democratic institutions contain a crisis transparently and irreversibly, investors update expectations about long-run institutional quality
- The four-phase dynamic — shock, uncertainty, turning point, revaluation — emerged from the data and maps precisely onto the political timeline
- **First statistically significant day: February 19, 2025** — when constitutional certainty was established, not at the shock
- **Full-period ATT** $\approx 24\%$ — the capitalized value of removing the Yoon political risk premium
- "Markets noticed, and they rewarded it"

Appendix 1: Pre-Treatment Fit (ASCM)

Metric	Value
RMSE	0.0233
MAE	0.0184
Bias	0.0000
SD	0.0234
Post-mean ATT	0.2427
SNR	10.4
λ	0.00395
L2 (scaled)	0.352

- Bias ≈ 0: synthetic neither overshoots nor undershoots Korea in pre-period
- RMSE of 0.023 $\approx \pm 2.3\%$ in price terms — within normal daily noise
- **SNR = 10.4**: post-treatment ATT is *10*× the pre-treatment noise level
- Moderate L2 driven by August 2024 spike (Yoon–Assembly confrontation + Yen carry trade unwind) — *consistent with* the political risk mechanism

Appendix 1: Pre-Treatment Fit (ASCM)

Metric	Value
RMSE	0.0233
MAE	0.0184
Bias	0.0000
SD	0.0234
Post-mean ATT	0.2427
SNR	10.4
λ	0.00395
L2 (scaled)	0.352

- Bias ≈ 0: synthetic neither overshoots nor undershoots Korea in pre-period
- RMSE of 0.023 $\approx \pm 2.3\%$ in price terms — within normal daily noise
- **SNR = 10.4**: post-treatment ATT is *10*× the pre-treatment noise level
- Moderate L2 driven by August 2024 spike (Yoon–Assembly confrontation + Yen carry trade unwind) — *consistent with* the political risk mechanism

Appendix 1: Pre-Treatment Fit (ASCM)

Metric	Value
RMSE	0.0233
MAE	0.0184
Bias	0.0000
SD	0.0234
Post-mean ATT	0.2427
SNR	10.4
λ	0.00395
L2 (scaled)	0.352

- Bias ≈ 0: synthetic neither overshoots nor undershoots Korea in pre-period
- RMSE of 0.023 $\approx \pm 2.3\%$ in price terms — within normal daily noise
- **SNR = 10.4**: post-treatment ATT is *10*× the pre-treatment noise level
- Moderate L2 driven by August 2024 spike (Yoon–Assembly confrontation + Yen carry trade unwind) — *consistent with* the political risk mechanism

Appendix 1: Pre-Treatment Fit (ASCM)

Metric	Value
RMSE	0.0233
MAE	0.0184
Bias	0.0000
SD	0.0234
Post-mean ATT	0.2427
SNR	10.4
λ	0.00395
L2 (scaled)	0.352

- Bias ≈ 0: synthetic neither overshoots nor undershoots Korea in pre-period
- RMSE of 0.023 $\approx \pm 2.3\%$ in price terms — within normal daily noise
- **SNR = 10.4**: post-treatment ATT is *10*× the pre-treatment noise level
- Moderate L2 driven by August 2024 spike (Yoon–Assembly confrontation + Yen carry trade unwind) — *consistent with* the political risk mechanism

Appendix 2: ATT by Post-Treatment Window

Window	N	ATT	CI lower	CI upper	
1 week (5 days)	5	-0.053	-0.104	-0.001	$*$
2 weeks (10 days)	10	-0.033	-0.085	0.018	
1 month (21 days)	21	-0.030	-0.082	0.021	
6 weeks (30 days)	30	-0.022	-0.073	0.030	
2 months (42 days)	42	0.016	-0.035	0.069	
3 months (63 days)	63	0.063	0.012	0.116	$*$
Full post-period	207	0.243	0.192	0.296	$***$

Jackknife+ 95% CI. $^{*}p < 0.05$, $^{***}p < 0.01$. ATT in log price units ($\times 100 \approx \%$ impact).

- CI excludes zero on **178 of 207** post-treatment days (86%)
- Full-period ATT $\approx 24\%$ cumulative revaluation

Appendix 2: ATT by Post-Treatment Window

Window	N	ATT	CI lower	CI upper	
1 week (5 days)	5	-0.053	-0.104	-0.001	$*$
2 weeks (10 days)	10	-0.033	-0.085	0.018	
1 month (21 days)	21	-0.030	-0.082	0.021	
6 weeks (30 days)	30	-0.022	-0.073	0.030	
2 months (42 days)	42	0.016	-0.035	0.069	
3 months (63 days)	63	0.063	0.012	0.116	$*$
Full post-period	207	0.243	0.192	0.296	$***$

Jackknife+ 95% CI. $^{*}p < 0.05$, $^{***}p < 0.01$. ATT in log price units ($\times 100 \approx \%$ impact).

- CI excludes zero on **178 of 207** post-treatment days (86%)
- Full-period ATT $\approx 24\%$ cumulative revaluation

Appendix 3: Dynamic Treatment Effects — Four Phases

- **Phase 1 — Shock (Week 1):** ATT $= -0.053^{*}$. Martial law $\Rightarrow$ uncertainty spike, negative returns. 0% of days positive.
- **Phase 2 — Uncertainty (Weeks 2–6):** ATT ≈ -0.02 to -0.03, *insignificant*.
- **Phase 3 — Turning point ($\sim$Month 2):** ATT $\approx +0.016$, *insignificant*. Turns positive as Yoon arrested (Jan 15).
- **Phase 4 — Revaluation (Month 3+):** ATT $= +0.063$ to $+0.243^{***}$. First significant: February 19, 2025.

Daily ATT (gap plot) with jackknife+ 95% CI

Appendix 3: Dynamic Treatment Effects — Four Phases

- **Phase 1 — Shock (Week 1):** ATT $= -0.053^{*}$. Martial law $\Rightarrow$ uncertainty spike, negative returns. 0% of days positive.
- **Phase 2 — Uncertainty (Weeks 2–6):** ATT ≈ -0.02 to -0.03, *insignificant*.
- **Phase 3 — Turning point ($\sim$Month 2):** ATT $\approx +0.016$, *insignificant*. Turns positive as Yoon arrested (Jan 15).
- **Phase 4 — Revaluation (Month 3+):** ATT $= +0.063$ to $+0.243^{***}$. First significant: February 19, 2025.

Daily ATT (gap plot) with jackknife+ 95% CI

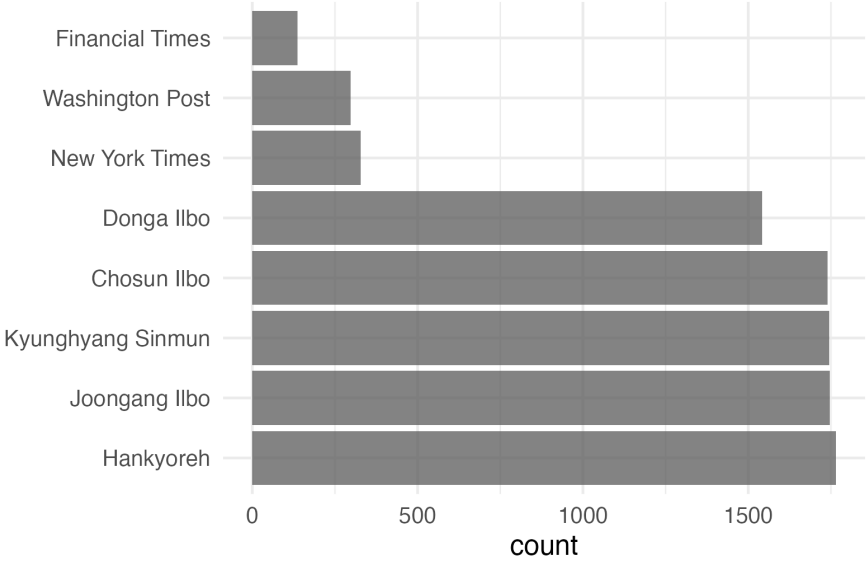
Appendix 3: Dynamic Treatment Effects — Four Phases

- **Phase 1 — Shock (Week 1):** ATT $= -0.053^{*}$. Martial law $\Rightarrow$ uncertainty spike, negative returns. 0% of days positive.
- **Phase 2 — Uncertainty (Weeks 2–6):** ATT ≈ -0.02 to -0.03, *insignificant*.
- **Phase 3 — Turning point ($\sim$Month 2):** ATT $\approx +0.016$, *insignificant*. Turns positive as Yoon arrested (Jan 15).
- **Phase 4 — Revaluation (Month 3+):** ATT $= +0.063$ to $+0.243^{***}$. First significant: February 19, 2025.

Daily ATT (gap plot) with jackknife+ 95% CI

Appendix 3: Dynamic Treatment Effects — Four Phases

- **Phase 1 — Shock (Week 1):** ATT $= -0.053^{*}$. Martial law $\Rightarrow$ uncertainty spike, negative returns. 0% of days positive.
- **Phase 2 — Uncertainty (Weeks 2–6):** ATT ≈ -0.02 to -0.03, *insignificant*.
- **Phase 3 — Turning point ($\sim$Month 2):** ATT $\approx +0.016$, *insignificant*. Turns positive as Yoon arrested (Jan 15).
- **Phase 4 — Revaluation (Month 3+):** ATT $= +0.063$ to $+0.243^{***}$. First significant: February 19, 2025.

Daily ATT (gap plot) with jackknife+ 95% CI

Appendix 4: News Outlet Makeup